

# Data Analytics Internship — Level 3

## Task 1: Restaurant Reviews

**Data note:** This dataset does not contain free-text customer reviews. It includes a categorical **Rating Text** field (Excellent / Very Good / Good / Average / Poor / Not Rated) instead. The analysis below treats these rating labels as a proxy for review sentiment and "review length," since no actual review text is available to mine for keywords.

## Objective

This report analyzes restaurant "review" sentiment (via the Rating Text labels) to identify the most common positive and negative keywords, calculates the average length of these labels, and explores whether there's a relationship between label length and the underlying numeric rating — addressing the Level 3, Task 1 brief as closely as the available data allows.

## Dataset Overview

| Metric                          | Value |
|---------------------------------|-------|
| Total restaurant records        | 9,551 |
| Restaurants with a rating label | 7,403 |
| Restaurants with "Not rated"    | 2,148 |

## 1. Sentiment Distribution

Rating Text labels were grouped into sentiment buckets: Excellent/Very Good/Good → Positive, Average → Neutral, Poor → Negative, and Not Rated as its own category.

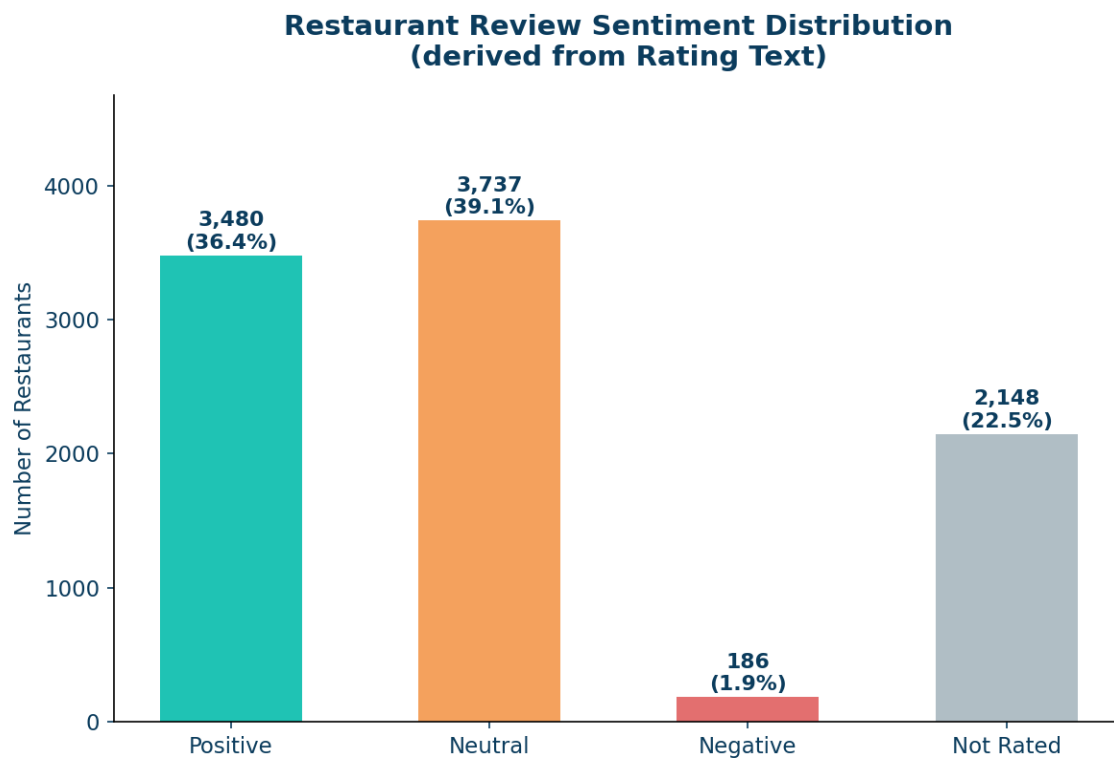


Figure 1: Distribution of restaurants across sentiment buckets.

## 2. Most Common Positive & Negative Keywords

Since the labels themselves function as the only available "keywords," their frequencies are shown below. "Good" is by far the most common positive descriptor, while "Poor" is the only negative label and is comparatively rare.

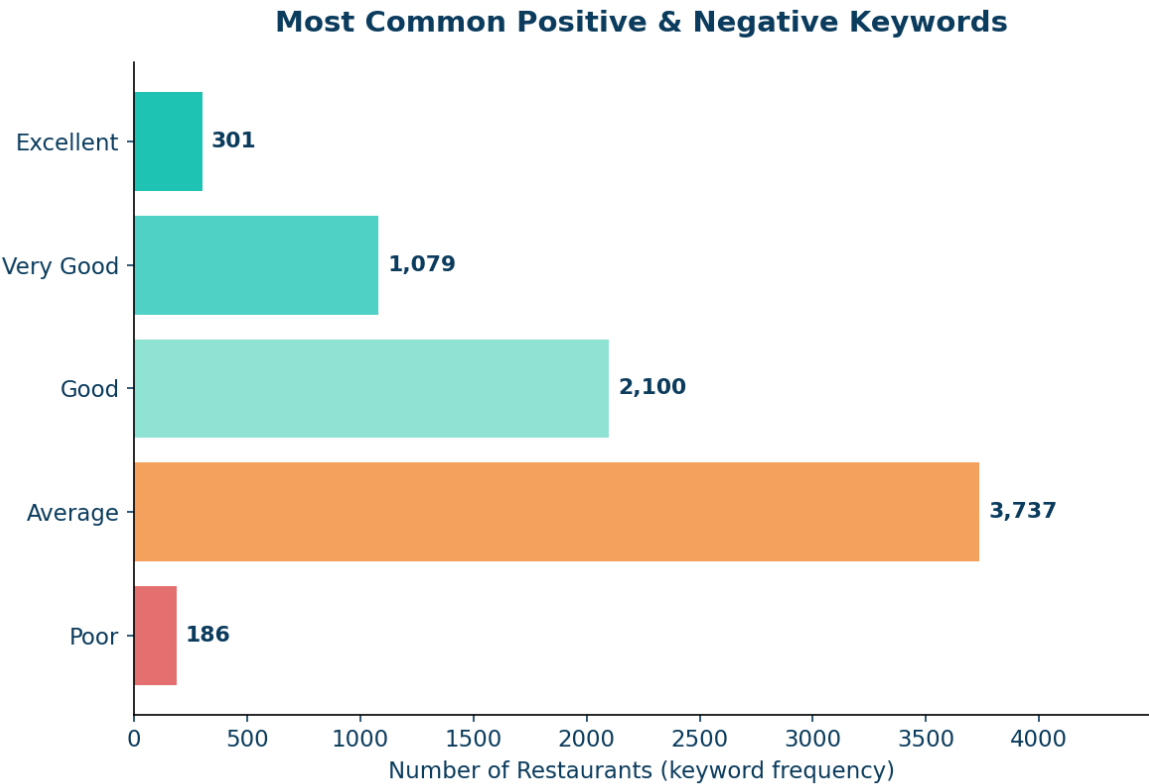


Figure 2: Frequency of each rating-text keyword.

| Keyword   | Sentiment | Restaurant Count |
|-----------|-----------|------------------|
| Good      | Positive  | 2,100            |
| Very Good | Positive  | 1,079            |
| Excellent | Positive  | 301              |
| Average   | Neutral   | 3,737            |
| Poor      | Negative  | 186              |

### 3. Label Length vs. Rating

As a proxy for "review length," the character length of each Rating Text label was compared against the restaurant's numeric Aggregate Rating. The overall correlation is weak-to-moderate ( $r = 0.192$ ), which is expected: longer labels like "Excellent" and "Very Good" (9 characters) correspond to higher ratings, while "Good" and "Poor" are both short (4 characters) despite representing opposite sentiments — so label length alone is not a reliable predictor of rating.

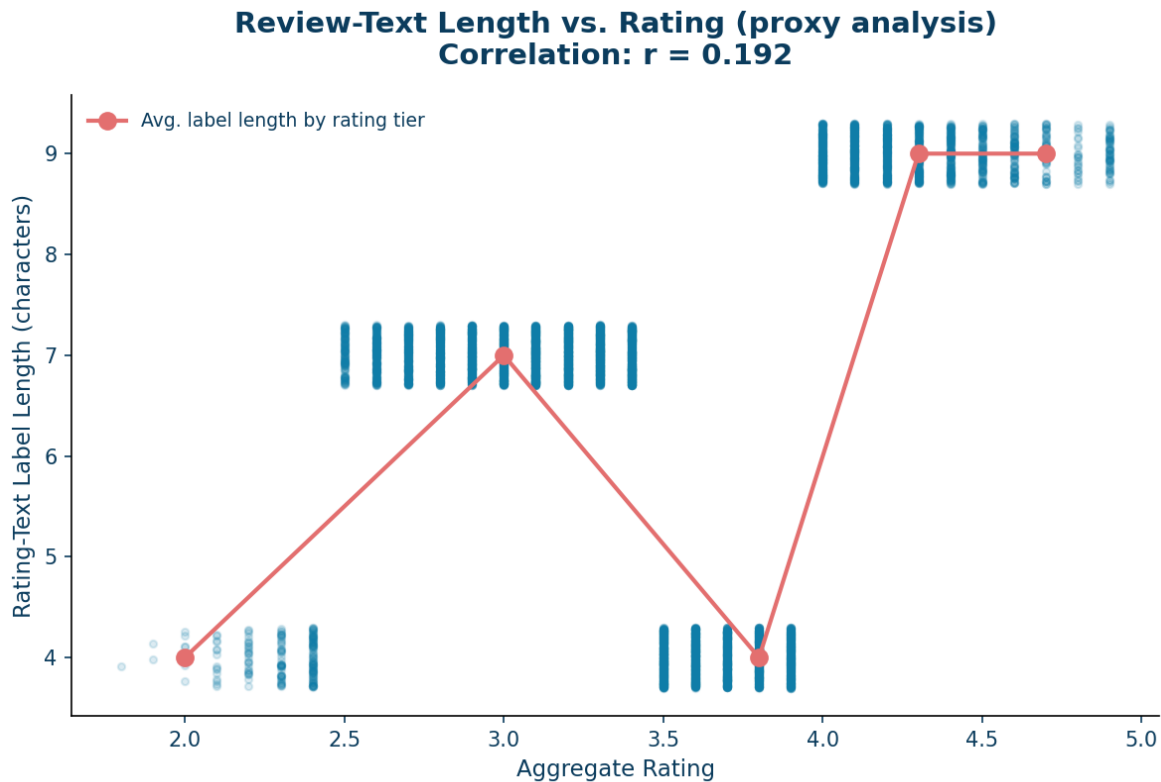


Figure 3: Rating-text label length plotted against aggregate rating, with tier averages overlaid.

## Conclusion

Because this dataset lacks genuine free-text reviews, the findings here are best read as a proxy analysis of rating sentiment rather than true text-mining results. "Good" and "Average" dominate the sentiment landscape, negative feedback ("Poor") is rare, and there is no strong, reliable relationship between label length and rating. For a genuine keyword and review-length analysis, a dataset containing actual customer review text would be needed.